W6 & W7 HARD

SCENARIO PRACTICE

Target Tracking + WMSN + Coverage + MWSN + Human Sensing + UAV/FANET

Long questions, hard cases, traps and exam decision maps

WHAT THIS PDF TRAINS

- Long 8-mark scenario questions
- Target tracking identification: Push / Poll / Guided
- WMSN CS vs SS + topology-management reasoning
- Coverage, connectivity, Rc >= 2Rs and OGDC
- Stationary vs Mobile WSN + Self-CHOP
- Human-centric sensing + DTN
- UAV Star/Mesh + FANET gateway selection and trajectory control

Priority First

PPG/CTF -> CS/SS -> CCL -> APB -> 2Rs -> OGDC -> CHOP

TOD/EPP -> LHAE -> Star/Mesh -> FANET -> DSO2

Scenario 1: Continuous Tiger Tracking

Long 8-mark pattern - identify, justify, explain

SCENARIO

A forest-monitoring WSN tracks a moving tiger. Sensor nodes calculate the tiger's position and periodically send the location to the sink even when the sink has not requested an update.

QUESTION

Which target-tracking formulation is used and what structure is commonly associated with it?

Best answer: Push-Based Tracking

- Sensor nodes compute the position of the target.
- Nodes periodically notify the sink.
- Reports are pushed automatically rather than waiting for a query.
- A cluster structure is commonly used in this formulation.

MEMORY

Push -> Cluster.

Scenario 2: Query-Based Intruder Tracking

Long 8-mark pattern - identify, justify, explain

SCENARIO

A security WSN detects an intruder but does not continuously report the target position. Nodes register the target's presence and send target data only when the sink issues a query.

QUESTION

Which tracking formulation is being used?

Best answer: Poll-Based Tracking

- Nodes register the presence of the target.
- Data reports are transmitted only when a query must be answered.
- This reduces unnecessary reporting compared with periodic push reporting.
- A tree structure is often used.

MEMORY

Poll -> Tree.

Scenario 3: Beacon-Guided Tracker

Long 8-mark pattern - identify, justify, explain

SCENARIO

Several sensor nodes act as beacons and define a path toward a moving target. A tracker follows the path to intercept the target.

QUESTION

Which tracking formulation is described?

Best answer: Guided Tracking

- Beacon nodes define the trajectory toward the target.
- The tracker follows the beacon trail.
- The goal is to intercept the target.
- A face structure is often used.

MEMORY

Guided -> Face.

Scenario 4: Full Tracking Comparison

Long 8-mark pattern - identify, justify, explain

SCENARIO

System A periodically reports target location. System B sends reports only after a query. System C uses beacon nodes to guide a tracker toward the target.

QUESTION

Identify all three methods and the common structure associated with each.

Best answer: Push-Cluster, Poll-Tree, Guided-Face

- System A -> Push-Based -> Cluster.
- System B -> Poll-Based -> Tree.
- System C -> Guided -> Face.
- The key difference is periodic reporting vs query-driven reporting vs physical guidance.

MEMORY

PPG -> CTF = Push-Cluster | Poll-Tree | Guided-Face.

Scenario 5: Precise Wildlife Sensing

Long 8-mark pattern - identify, justify, explain

SCENARIO

A wildlife monitoring application needs video and audio so that animals can be identified precisely. Temperature-only sensors cannot provide enough detail.

QUESTION

Which WMSN node type should be used?

Best answer: Camera Sensor (CS)

- Camera Sensor nodes capture multimedia such as video and audio.
- They can also handle scalar information.
- They are more resource hungry and comparatively expensive.
- They have directional sensing range.
- WMSNs generally use fewer CS nodes than SS nodes.

MEMORY

Camera = Costly + Clear.

Scenario 6: Low-Cost Environmental Monitoring

Long 8-mark pattern - identify, justify, explain

SCENARIO

An environmental network only needs temperature, light and vibration readings. The designer wants low-cost nodes and does not need multimedia.

QUESTION

Which WMSN node type is more appropriate?

Best answer: Scalar Sensor (SS)

- Scalar Sensor nodes sense data such as temperature, light and vibration.
- They are low cost.
- They have omni-directional sensing range.
- A WMSN normally contains a larger number of SS nodes.

MEMORY

Scalar = Simple.

Scenario 7: Video Network Battery Drain

Long 8-mark pattern - identify, justify, explain

SCENARIO

A WMSN sends video from an event area to a control center. The video needs large bandwidth and drains battery quickly. The designer must still sense the event and deliver the video successfully.

QUESTION

Which three topology-management objectives are most important?

Best answer: CCL - Coverage, Connectivity, Lifetime

- Coverage ensures the event is sensed quickly.
- Connectivity ensures video data can travel from the event area to the control center.
- Network lifetime is important because multimedia transmission consumes high energy.
- WMSN topology management must balance these three goals.

MEMORY

CCL = Coverage - Connectivity - Lifetime.

Scenario 8: Mixed Camera and Scalar Design

Long 8-mark pattern - identify, justify, explain

SCENARIO

A wildlife reserve uses a few camera sensors for precise information and many scalar sensors for basic environmental sensing. The goal is accurate monitoring without exhausting energy too quickly.

QUESTION

Why is a mixed CS + SS deployment useful?

Best answer: Precision + Low-Cost Coverage + Longer Lifetime

- CS nodes provide precise multimedia information.
- SS nodes provide lower-cost scalar sensing.
- Using fewer CS nodes reduces expensive resource-heavy sensing.
- Many SS nodes provide broader simple sensing.
- The combined deployment can improve sensing and prolong network lifetime.

MEMORY

Few CS + many SS = better sensing with controlled resource use.

Scenario 9: Whole Field Monitoring

Long 8-mark pattern - identify, justify, explain

SCENARIO

Sensors are deployed over an agricultural field and every part of the field must remain monitored.

QUESTION

Which static WSN coverage problem is this?

Best answer: Area Coverage

- Area coverage concerns the entire monitoring region.
- Its objective is to ensure that the sensing field is satisfactorily covered.
- It may be implemented using energy-efficient or connected random coverage approaches.

MEMORY

Area = All area.

Scenario 10: Selected Machine Locations

Long 8-mark pattern - identify, justify, explain

SCENARIO

A factory has five high-risk machines. Only these exact machine locations must always be sensed; the spaces between them are not important.

QUESTION

Which coverage problem fits?

Best answer: Point Coverage

- Point coverage focuses on a set of selected points.
- Random point coverage distributes sensors so every required point remains covered.
- Deterministic point coverage does the same using planned placement.

MEMORY

Point = Particular points.

Scenario 11: Border Crossing Detection

Long 8-mark pattern - identify, justify, explain

SCENARIO

Sensors are deployed along a restricted boundary. Any intruder crossing the boundary must be detected.

QUESTION

Which coverage problem is represented?

Best answer: Barrier Coverage

- Barrier coverage focuses on detecting crossings through a monitored barrier.
- 1-barrier coverage means at least one sensor covers the crossing.
- 2-barrier coverage means at least two sensors cover it.
- K-barrier coverage means at least K sensors cover the crossing.

MEMORY

Barrier = Boundary crossing.

Scenario 12: Three-Sensor Border Requirement

Long 8-mark pattern - identify, justify, explain

SCENARIO

A military boundary requires every crossing to be detected by at least three sensors.

QUESTION

What type of barrier coverage is required?

Best answer: 3-Barrier Coverage

- The general rule is K-barrier coverage.
- A crossing is covered by at least K sensors.
- Here K = 3, so the requirement is 3-barrier coverage.

MEMORY

K-barrier = at least K sensors.

Scenario 13: Sense But Cannot Send

Long 8-mark pattern - identify, justify, explain

SCENARIO

A sensor successfully detects an event, but because neighboring links are broken, its information cannot reach the sink node.

QUESTION

Which property is present and which property has failed?

Best answer: Coverage exists, Connectivity fails

- Coverage means the event or area is sensed.
- Connectivity means nodes are connected so sensed data can reach the sink.
- The event is detected, so coverage exists.
- The data cannot reach the sink, so connectivity fails.

MEMORY

Coverage = Sense | Connectivity = Send.

Scenario 14: Range Rule Satisfied

Long 8-mark pattern - identify, justify, explain

SCENARIO

A sensor network has sensing range Rs = 20 m and communication range Rc = 45 m.

QUESTION

Using the slide rule, can coverage imply connectivity?

Best answer: Yes, because Rc >= 2Rs

- 2Rs = 40 m.
- Rc = 45 m.
- 45 >= 40, so the sufficient condition is satisfied.
- Therefore, according to the slide rule, coverage implies connectivity.

MEMORY

Rc >= 2Rs -> Coverage => Connectivity.

Scenario 15: Range Rule Not Satisfied

Long 8-mark pattern - identify, justify, explain

SCENARIO

A network has Rs = 30 m and Rc = 50 m.

QUESTION

Does the sufficient condition for coverage implying connectivity hold?

Best answer: No

- 2Rs = 60 m.
- Rc = 50 m.
- 50 < 60, so Rc >= 2Rs is not satisfied.
- Therefore this rule alone cannot guarantee connectivity from coverage.

MEMORY

Always calculate 2Rs first.

Scenario 16: Central Base Station Decides

Long 8-mark pattern - identify, justify, explain

SCENARIO

Every sensor sends its information to one base station. The base station computes a global map and decides which nodes should remain active.

QUESTION

Which coverage algorithm type is being used?

Best answer: Centralized Algorithm

- Data is collected at a central point.
- The global map is computed centrally.
- Decision-making depends on the central controller.

MEMORY

Centralized = Central point + global map.

Scenario 17: Neighbor-Based Coverage Decision

Long 8-mark pattern - identify, justify, explain

SCENARIO

Each sensor makes a coverage decision using only information exchanged with neighboring nodes.

QUESTION

Which algorithm type is this?

Best answer: Distributed Algorithm

- Nodes communicate with their neighbors.
- Decisions are made without collecting all data at one central point.
- The processing responsibility is distributed among nodes.

MEMORY

Distributed = Neighbors decide.

Scenario 18: Only a Subset Participates

Long 8-mark pattern - identify, justify, explain

SCENARIO

Only selected sensors participate in sensing, communication and computation while the rest remain inactive.

QUESTION

Which coverage algorithm type is described?

Best answer: Localized Algorithm

- Localized algorithms are a special type of distributed method.
- Only a subset of nodes participates in sensing, communication and computation.
- This can reduce unnecessary resource use.

MEMORY

Localized = Limited subset participates.

Scenario 19: OGDC Energy Saving

Long 8-mark pattern - identify, justify, explain

SCENARIO

A dense WSN has many overlapping sensors. The designer wants to maintain coverage while keeping redundant sensors asleep to save energy.

QUESTION

Which algorithm from W6&7 should be used and how does it work?

Best answer: OGDC - Optimal Geographical Density Control

- A starting node becomes active.
- It broadcasts its position and ideal/preferred direction.
- A node closest to the ideal distance and angle becomes active.
- Nodes continue to cover uncovered crossings with minimum overlap.
- A redundant node can sleep when its coverage area is fully covered.

MEMORY

SICCS = Start - Ideal direction - Closest - Cover - Sleep.

Scenario 20: OGDC Starting Node

Long 8-mark pattern - identify, justify, explain

SCENARIO

At the beginning of OGDC, nodes wait for a random back-off time. If a node hears nothing from neighbors, it declares itself as the starting node and announces its position and direction.

QUESTION

Which OGDC phase is described?

Best answer: Starting-Node Selection

- Nodes may voluntarily participate with probability p.
- They choose a random back-off time.
- A node that hears nothing can declare itself the starting node.
- The starting node announces position and preferred direction.

MEMORY

Starting node = random back-off + no neighbor announcement heard.

Scenario 21: Static Failure Splits Network

Long 8-mark pattern - identify, justify, explain

SCENARIO

A WSN uses fixed sensors. One critical sensor fails and the network becomes partitioned into two disconnected parts.

QUESTION

Which network limitation does this show?

Best answer: Stationary WSN Disadvantage

- Stationary sensors are static.
- A node failure may partition the network.
- The topology cannot change automatically.
- This is one reason mobility can be useful.

MEMORY

Stationary = Simple but Stuck.

Scenario 22: Dynamic Mobile Recovery

Long 8-mark pattern - identify, justify, explain

SCENARIO

Sensors are able to move and reorganize after topology changes. The network is expected to reconfigure and recover more dynamically.

QUESTION

Which concept is more suitable?

Best answer: Mobile Wireless Sensor Network (MWSN)

- MWSN introduces mobility into sensor networking.
- It is related to MANET-style infrastructure-less wireless networking.
- Dynamic topology changes are possible.
- MANET self-organizing behavior is summarized by Self-CHOP.

MEMORY

Mobile -> Self-CHOP.

Scenario 23: Self-CHOP Identification

Long 8-mark pattern - identify, justify, explain

SCENARIO

A mobile sensor network can set itself up, recover from failures, improve operation and defend itself.

QUESTION

Name the four MANET-style properties.

Best answer: Self-CHOP

- Self-Configure.
- Self-Heal.
- Self-Optimize.
- Self-Protect.

MEMORY

CHOP = Configure - Heal - Optimize - Protect.

Scenario 24: Smartphone City Sensing

Long 8-mark pattern - identify, justify, explain

SCENARIO

Citizens carry smartphones with accelerometers and gyroscopes. Their devices collect readings while they move around the city.

QUESTION

Which sensing idea is represented?

Best answer: Human-Centric Sensing

- Humans carry sensor-equipped devices.
- Smartphones and PDAs contain sensors such as accelerometers and gyroscopes.
- Embedded sensors record readings while people move.
- The readings can then be transmitted.
- Related paradigms include participatory, people-centric and opportunistic sensing.

MEMORY

Humans + Phones + Moving Sensors.

Scenario 25: Three Human Roles

Long 8-mark pattern - identify, justify, explain

SCENARIO

Person A is a patient whose health is measured. Person B uses a phone sensor to measure pollution. Person C posts traffic information online without directly using a sensor.

QUESTION

Identify the three human-centric sensing roles.

Best answer: Target, Operator, Data Source

- Person A -> Sensing Target.
- Person B -> Sensor Operator.
- Person C -> Data Source.
- These roles are not necessarily mutually exclusive.

MEMORY

TOD = Target - Operator - Data Source.

Scenario 26: Human Sensing Challenges

Long 8-mark pattern - identify, justify, explain

SCENARIO

A smartphone sensing campaign suffers from battery drain, difficulty choosing reliable participants and concern about personal information.

QUESTION

Which three challenges are shown?

Best answer: EPP

- Energy of devices.
- Participant selection.
- Privacy of users.

MEMORY

EPP = Energy - Participant - Privacy.

Scenario 27: Participatory CO2 Sensing

Long 8-mark pattern - identify, justify, explain

SCENARIO

Citizens use their own devices to collect CO2 values. The readings include geotags and timestamps and are shared with common users.

QUESTION

Which sensing paradigm is this?

Best answer: Participatory Sensing

- It is distributed sensing by devices carried by humans.
- Its goal is not only data collection but also public access and knowledge sharing.
- Data can include quantitative information such as CO2 level.
- Geotagged location and timestamp can support authenticity.
- The slide credits Burke et al., 2006.

MEMORY

People + Phones + Share = Participatory Sensing.

Scenario 28: Intermittent Remote Network

Long 8-mark pattern - identify, justify, explain

SCENARIO

A remote sensor network often has no complete end-to-end path. Delay is high, data rates are asymmetric and the channel is error prone.

QUESTION

Which network category fits this scenario?

Best answer: Delay Tolerant Network (DTN)

- End-to-end communication paths may be unavailable.
- Latency can be high.
- Data rates can be asymmetric.
- Channels may be erroneous.
- Data MULE approaches are related to such delay-tolerant WSN settings.

MEMORY

LHAE = Lack of path - High latency - Asymmetry - Errors.

Scenario 29: Disaster UAV Deployment

Long 8-mark pattern - identify, justify, explain

SCENARIO

Several UAVs are rapidly deployed over a disaster zone. The network needs wide coverage, adaptive routing and easy mission reconfiguration.

QUESTION

Which UAV-network features support this requirement?

Best answer: Flexible, Adaptive, Large-Coverage UAV Networking

- UAV networks can use Mesh or Star topology.
- Deployment and new service management can be flexible using SDN.
- Routing should be adaptive.
- UAV networks can support multi-tasking.
- They offer large coverage and can be reconfigured for varying missions.

MEMORY

Think: Flexible + Adaptive + Large + Reconfigurable.

Scenario 30: Single vs Multi-UAV Mission

Long 8-mark pattern - identify, justify, explain

SCENARIO

Mission A wants simple control and can tolerate limited scalability. Mission B needs fast completion, high scalability and better survival when a UAV fails.

QUESTION

Which system should each mission use?

Best answer: A = Single UAV, B = Multi-UAV

- Single UAV has lower control complexity but limited scalability and poorer survivability.
- Multi-UAV has higher scalability and survivability.
- Multi-UAV missions can be faster.
- Multi-UAV systems cost more and have higher control complexity.

MEMORY

Single = Simple but weak | Multi = Complex but strong.

Scenario 31: Central UAV Controller

Long 8-mark pattern - identify, justify, explain

SCENARIO

Every UAV communicates through a central controller. Nodes do not relay traffic for one another and communication is single-hop to the center.

QUESTION

Which UAV topology is this?

Best answer: Star Topology

- Star is point-to-point.
- A central control point is present.
- It is infrastructure-based.
- It is not self-configuring.
- Communication is single-hop to the central point.

MEMORY

Star = Center.

Scenario 32: Autonomous UAV Relay

Long 8-mark pattern - identify, justify, explain

SCENARIO

UAVs relay packets for one another, use multi-hop links and may operate without a central control point.

QUESTION

Which topology is described?

Best answer: Mesh Topology

- Mesh supports multipoint-to-multipoint communication.
- It may be infrastructure-based or ad hoc.
- Ad hoc mesh can have no central controller.
- It is self-configuring.
- Nodes can relay traffic for other nodes.

MEMORY

Mesh = Members help Members.

Scenario 33: UAV + Ground Cooperation

Long 8-mark pattern - identify, justify, explain

SCENARIO

A group of UAVs communicates with other UAVs, ground sensors, a ground station and vehicles in a VANET.

QUESTION

Which network concept is being used?

Best answer: FANET - Flying Ad Hoc Network

- FANET supports inter-plane and intra-plane communication.
- It can communicate with ground stations and ground sensors.
- It can also interact with VANETs.
- Ad hoc FANET A2A links may use IEEE 802.15.4/XBee-PRO or Wi-Fi/IEEE 802.11.

MEMORY

FANET = Flying Ad Hoc Network.

Scenario 34: UAV Data Muling

Long 8-mark pattern - identify, justify, explain

SCENARIO

A UAV flies over a stationary WSN, collects sensor data and physically carries the data toward another network location.

QUESTION

Which FANET cooperation is demonstrated?

Best answer: UAV-WSN Data Muling

- UAV-WSN link-up can support collaborative sensing.
- It can also support data muling.
- The UAV acts as a mobile carrier for sensor information.

MEMORY

UAV + WSN + carry data = Data Muling.

Scenario 35: UAV Helps VANET

Long 8-mark pattern - identify, justify, explain

SCENARIO

UAVs cooperate with a vehicular network to provide visual guidance and improve communication coverage.

QUESTION

Which FANET cooperation is this?

Best answer: UAV-VANET Cooperation

- UAV-VANET link-up can support visual guidance.
- It can support data muling.
- It can help enhance coverage.

MEMORY

UAV-VANET = guidance + muling + coverage.

Scenario 36: FANET Gateway Selection

Long 8-mark pattern - identify, justify, explain

SCENARIO

A large UAV communication area is divided into smaller sub-areas. One stable UAV should act as gateway in each region, and UAVs maintain limited neighborhood information.

QUESTION

Explain the gateway-selection logic.

Best answer: Divide -> Stable Node -> Optimize -> 2-Hop

- Divide the entire UAV coverage area into sub-areas.
- The sub-areas collectively cover the communication region.
- Adjust sub-area sizes dynamically.
- Select the most stable node in a sub-area.
- Optimize partition/topology parameters.
- Each UAV acquires information about UAVs within 2 hops.

MEMORY

DSO2 = Divide - Stable - Optimize - 2 hops.

Scenario 37: Congested UAV Queue

Long 8-mark pattern - identify, justify, explain

SCENARIO

A UAV has queue occupancy above a threshold. The busy communication link causes delay and congestion.

QUESTION

What can trajectory control do?

Best answer: Change UAV Trajectory

- Queue occupancy above threshold indicates congestion.
- Congestion causes communication delay.
- The control station can instruct UAVs to change trajectory centers.
- The command depends on traffic at busy communication links.
- Trajectory radius can also be changed to improve coverage.

MEMORY

Busy Queue -> Move UAV.

Hard Integrated Design 1

Smart Border Surveillance - combine multiple W6&7 concepts

SCENARIO

A border system needs every crossing detected by two sensors, camera confirmation, vibration sensing, long network lifetime and UAVs that relay data from remote sensor areas.

Crossing detection	->	Barrier Coverage
Two sensors per crossing	->	2-Barrier Coverage
Visual confirmation	->	Camera Sensor (CS)
Vibration sensing	->	Scalar Sensor (SS)
Coverage/connectivity/lifetime	->	WMSN CCL

DESIGN MEMORY

Barrier + CS/SS + CCL + FANET

Hard Integrated Design 2

Wildlife Monitoring - tracking + sensing + human + UAV

SCENARIO

A reserve needs periodic tiger-location reports, camera images, low-cost environmental sensing, researchers carrying smartphones and UAVs collecting data from remote sensor nodes.

Periodic target reports	->	**Push-Based Tracking**
Images/video	->	**Camera Sensor**
Temperature/light/vibration	->	**Scalar Sensor**
Researchers with smartphones	->	**Human-Centric Sensing**

FAST IDENTIFY

Push + CS + SS + Human-Centric + Data Muling

Hard Trap Questions

True/False + correction

1. Push-based tracking normally uses Tree structure.

FALSE - Push uses Cluster.

2. Poll-based sends reports periodically without query.

FALSE - Poll waits for query.

3. Guided tracking may use beacon nodes.

TRUE.

4. Scalar sensors are more resource hungry than camera sensors.

FALSE - CS is resource hungry.

5. Coverage means data can reach the sink.

FALSE - that is Connectivity.

6. Rc >= 2Rs is a sufficient condition used in the slides.

TRUE.

7. Point coverage means covering the whole field.

FALSE - selected points.

8. K-barrier means at least K sensors cover a crossing.

TRUE.

9. OGDC keeps all redundant nodes active.

FALSE - redundant nodes sleep.

10. Stationary WSN topology can always change automatically.

FALSE.

11. Self-CHOP contains Self-Heal.

TRUE.

12. A human can be a sensing target.

TRUE.

13. DTN requires a permanent end-to-end path.

FALSE.

14. Star topology is self-configuring multi-hop mesh.

FALSE.

15. Mesh nodes may relay traffic for other nodes.

TRUE.

16. FANET can communicate with VANET.

TRUE.

17. Gateway selection uses 2-hop UAV information.

TRUE.

Medium Practice - 26 Questions

Good 5-mark revision set

1. Explain Push-Based Target Tracking.
2. Explain Poll-Based Target Tracking.
3. Explain Guided Target Tracking.
4. Explain AID using PIR and Ultrasonic sensors.
5. Compare Camera Sensor and Scalar Sensor.
6. Explain WMSN Topology Management using CCL.
7. Explain Coverage vs Connectivity.
8. Explain Area Coverage.
9. Explain Point Coverage.
10. Explain Barrier Coverage.
11. Explain Rc >= 2Rs relation.
12. Compare Centralized, Distributed and Localized algorithms.
13. Explain OGDC Algorithm.
14. Stationary WSN: advantages and disadvantages.
15. Explain MWSN and Self-CHOP.
16. Explain Human-Centric Sensing.
17. Explain the three human roles.
18. Explain EPP challenges.
19. Explain Participatory Sensing.
20. Explain Delay Tolerant Networks.
21. List UAV Network features.
22. Compare Single UAV and Multi-UAV.
23. Compare Star and Mesh Topology.
24. Explain FANET.
25. Explain FANET Gateway Selection.
26. Explain UAV Trajectory Control.

Rapid Practice - 30 Questions

Try without looking

1. Name 3 target-tracking formulations.
2. Push uses which structure?
3. Poll uses which structure?
4. Guided uses which structure?
5. What does PIR detect?
6. What does Ultrasonic detect?
7. CS full form?
8. What does SS sense?
9. What is CCL?
10. What is Coverage?
11. What is Connectivity?
12. State Rc and Rs rule.
13. Name 3 coverage problems.
14. What is Point Coverage?
15. What is K-Barrier?
16. What is CDL?
17. OGDC full form?
18. What does redundant OGDC node do?
19. Give one Stationary WSN disadvantage.
20. What is Self-CHOP?
21. Name 3 human roles.
22. Name 3 human-centric challenges.
23. What is Participatory Sensing?
24. What is DTN?
25. Name 2 DTN characteristics.
26. Which has higher scalability: Single or Multi-UAV?
27. Which is self-configuring: Star or Mesh?
28. FANET full form?
29. What does UAV-WSN link support?
30. How many hops of neighbor info in FANET gateway selection?

Final Exam Strategy

How to solve an unseen W6&7 scenario quickly

MASTER DECISION MAP

Periodic target	->	Push	Neighbor compute	->	Distributed
Query-only target	->	Poll	Subset participates	->	Localized
Beacon trail	->	Guided	Redundant nodes sleep	->	OGDC
Video/audio costly	->	Camera Sensor	4 self-properties	->	Self-CHOP
Temp/light/vibration	->	Scalar Sensor	Humans carry devices	->	Human-Centric
Whole region	->	Area Coverage	No end-to-end path	->	DTN
Selected points	->	Point Coverage	Central UAV controller	->	Star
Crossing/border	->	Barrier Coverage	Multi-hop/self-config	->	Mesh
Sense, cannot send	->	Connectivity failure	Stable gateway + 2-hop	->	FANET DSO2
Rc >= 2Rs	->	Coverage => Connectivity			

8-MARK ANSWER FORMULA

- 1. Identify the exact model/algorithm.
- 2. Write a one-line definition.
- 3. Link scenario clues to the model.
- 4. Write 5-7 clean slide-based points.
- 5. Add a key relation/flow/comparison where relevant.
- 6. Finish with a one-line conclusion.

FINAL MEMORY

PPG/CTF -> CS/SS -> CCL -> APB -> 2Rs -> CDL -> OGDC -> CHOP -> TOD/EPP -> LHAE -> Star/Mesh -> FANET -> DSO2